DAY 14 | PRE-READ

When Data Gets Big

A light read before Day 14. About 15 minutes, no code required.

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Why read this?

Yesterday you cleaned a few rows of data with pandas on one computer. But what if a company has billions of rows, too many for any single machine? Tomorrow you meet Apache Spark, a tool that spreads the work across many computers.

1. Many hands make light work

Imagine counting every vote in a national election alone. It would take years. Instead, thousands of counting centres each count their own share at the same time, then send totals to a central office. Spark works exactly like this. A **driver** (the central office) splits the job, and many **executors** (the counting centres) work in parallel.

2. Plan first, then work

A good chef reads the whole recipe before starting, and notices they can chop all the vegetables at once. Spark does the same. When you describe what you want, Spark only writes it down. When you finally ask for a result, it looks at the whole plan, finds the quickest route, and then does the work. This is called **lazy evaluation**.

3. Raw, clean and ready

Layer	Kitchen version	Data version
Bronze	Groceries as delivered	Raw data exactly as received
Silver	Washed and chopped	Cleaned, fixed and de-duplicated
Gold	The finished dish	Summaries ready for reports

4. Matching up information

An orders list has customer numbers. A customer list has names and cities. To know which city ordered the most, you must match the two lists. This matching is called a **join**. Choosing how to treat orders with no matching customer is an important decision, and a common source of mistakes.

Key idea: after every join, count the rows. If the number surprises you, stop and find out why.

5. Think about these before class

- What large task at work could be split among many people working in parallel?
- Why might it help to plan a whole journey before starting to drive?
- If an order refers to a customer who is not in the customer list, should it appear in the report? Why?

Chiri mo tsumoreba yama to naru (塵も積もれば山となる): even dust, piled up, becomes a mountain